Lec 25

Bayes rule.

$$X \longrightarrow \boxed{Y|X} \xrightarrow{\;Y\;} \boxed{\text{Estimate}} \longrightarrow \hat{X} = \hat{x}(Y).$$

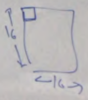
Prior distribution: $P_X(x)$ or $f_X(x)$

Likelihood distribution: $\underbrace{P_{Y|X}(y|x)}_{Y \text{ discrete}}$, $\underbrace{f_{Y|X}(y|x)}_{Y \text{ continuas}}$

Aposterior distribution: $P_{X|Y}(x|y)$ $f_{X|Y}(x|y)$

Example: $X \in \{0, 1, 2, \ldots 9\}$.

$$Y \in [0:255]^{16 \times 16}$$



X and Y are continuos

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{f_X(x)\, f_{Y|X}(y|x)}{\int_{-\infty}^{\infty} f_X(u)\, f_{Y|X}(y|u)\, du}$$

X and Y discrete

$$P_{X|Y}(x|y) = \frac{P_X(x)\, P_{Y|X}(y|x)}{\sum_{\tilde{x}\in\mathcal{X}} P_X(\tilde{x})\, P_{Y|X}(y|\tilde{x})}$$

$P(S=000 \mid R=101)$

$\boxed{P(R=101 \mid S=000)}$

X continuos & Y discrete

$$f_{X|Y}(x|y) = \frac{f_X(x)\, P_{Y|X}(y|x)}{\int_{-\infty}^{\infty} f_X(u)\, P_{Y|X}(y|u)\, du}$$

X discrete & Y continuos

$$P_{X|Y}(x|y) = \frac{P_X(x)\, f_{Y|X}(y|x)}{\sum_{u \in \mathcal{X}} P_X(u)\, f_{Y|X}(y|u)}$$

Independence

X, Y continuos are said to be independent

if $f_{X,Y}(x,y) = f_X(x)\, f_Y(y) \quad \forall x, y \in \mathbb{R}$

X, Y discrete are independent

$P_{X,Y}(x,y) = P_X(x)\, P_Y(y), \quad x \in \mathcal{X},\ y \in \mathcal{Y}$

X, Y independent

$\Longleftrightarrow \quad F_{X,Y}(x,y) = F_X(x)\, F_Y(y)$

$\Longleftrightarrow \quad P(X \in A, Y \in B) = P(X \in A)\, P(Y \in B)$

X, Y discrete.

Suppose $P_{X,Y}(x,y) = P_X(x)\,P_Y(y) \quad \forall x, y$

We'll show $F_{X,Y}(x,y) = F_X(x)\,F_Y(y)$

$$F_{X,Y}(x,y) = \sum_{\substack{u \le x \\ v \le y}} P_{X,Y}(u,v)$$

$$= \sum_{\substack{u \le x \\ v \le y}} P_X(u)\,P_Y(v)$$

$$= \left[\sum_{u \le x} P_X(u)\right]\left[\sum_{v \le y} P_Y(v)\right]$$

$$= F_X(x)\,F_Y(y)$$

We'll show That

$$F_{X,Y}(x,y) = F_X(x)\,F_Y(y) \quad \forall x, y \implies P_{X,Y}(x,y) = P_X(x)\,P_Y(y) \quad \forall x, y.$$

$P(x_1 < X \le x_2, y_1 < Y \le y_2)$

$$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1)$$

$P_{X,Y}(x, y)$

$$= P(x^- < X \le x,\ y^- < Y \le y)$$

$$= F_{X,Y}(x, y) - F_{X,Y}(x^-, y) - F_{X,Y}(x, y^-) + F_{X,Y}(x^-, y^-)$$

$$= F_X(x)\left[F_Y(y) - F_Y(y^-)\right] - F_X(x^-)F_Y(y) + F_X(x^-)F_Y(y^-)$$

$$= \left[F_X(x) - F_X(x^-)\right]\left[F_Y(y) - F_Y(y^-)\right]$$

$$= P_X(x)\,P_Y(y)$$

Continuos R.Vs

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$\Leftrightarrow \quad F_{X,Y}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f_{X,Y}(u,v)\,du\,dv$$

$$= \int_{-\infty}^{y}\int_{-\infty}^{x} f_X(u)\, f_Y(v)\,du\,dv$$

$$= \left(\int_{-\infty}^{y} f_Y(v)\,dv\right)\left(\int_{-\infty}^{x} f_X(u)\,du\right)$$

$$= F_Y(y)\, F_X(x)$$

$$P(X\in A, Y\in B) = \int_{y\in B}\int_{x\in A} f_{X,Y}(x,y)\,dx\,dy = \left(\int_{x\in A} f_X(x)\,dx\right)\left(\int_{y\in B} f_Y(y)\,dy\right)$$

Suppose $P(X\in A,\ Y\in B) = P(X\in A)\, P(Y\in B)$.

$\Longrightarrow \quad P_{X,Y}(x,y) = P_X(x)\, P_Y(y) \quad \forall x, y.$

[Assume $A=\{x\}$, $Y=\{y\}$].

$\Longrightarrow \quad f_{X,Y}(x,y)\,\delta_x\,\delta_y = f_X(x)\,\delta_x\, f_Y(y)\,\delta_y$

[Assume $A=[x, x+\delta_x]$, $B=[y, y+\delta_y]$]

If X, Y are independent

ⓐ $E[XY] = E[X]E[Y] \Leftrightarrow Cov(X,Y)=0$

$\Uparrow$

ⓑ $E[h(x)\,g(y)] = E[h(x)]\,E[g(y)]$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x)\,g(y)\,f_{X,Y}(x,y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x)\,g(y)\,f_X(x)\,f_Y(y)\,dx\,dy$$

$$= \left[\int_{-\infty}^{\infty} h(x)\,f_X(x)\,dx\right]\left[\int_{-\infty}^{\infty} g(y)\,f_Y(y)\,dy\right]$$

$$= E[h(x)]\,E[g(Y)]$$

ⓒ $Var(X+Y) = Var(X) + Var(Y)$

$$\left[\int_{y \in B} f_Y(y)\,dy\right] = P(X \in A)\,P(Y \in B)$$

Lec 25

Example:

$Cov(X,Y)=0 \not\Rightarrow$ X, Y are independent

$E[X|Y]=E[X] \not\Rightarrow$ X, Y are independent

Exam Qn 8

$$E[f(Y)g(X)|Y] = f(Y)E[g(X)|Y]. \text{ (True)}$$

$Var(X) = E[Var(X|Y)]$. Total law of variance

$$Var(\underbrace{E[X|Y]}_{Z}) = 0$$

$$\Rightarrow Z = E[Z]$$

$$E[X|Y] = E[E[X|Y]]$$

$$= E[X]$$

$\Rightarrow Cov(X,Y)=0.$

If $E[X|Y] = E[X]$
(mean independence).

Then

$$E[XY] = E\big[E[XY|Y]\big]$$

$$= E\big[Y\,E[X|Y]\big].$$

as $E[f(Y)g(X)|Y] = f(Y)E[g(X)|Y]$

$$= E\big[Y\,E[X]\big]$$

$$= E[X]\,E[Y]$$

$\Rightarrow \operatorname{Cov}(X,Y) = 0$

Qn

① Does X, Y independent

$\Rightarrow E[X|Y] = E[X]$? Yes.

$$E[X|Y=y] = \sum_{x\in\mathcal{X}} P_{X|Y}(x|y)\,x$$

$$= \sum_{x\in\mathcal{X}} P_X(x)\,x$$

$$= E[X]$$

①

$\mathrm{Cov}(X,Y)=0 \not\Rightarrow X, Y$ are independent.

doesn't imply

① $X \sim U[-1,1]$

$Y = X^2 \quad \mathrm{Cov}(X,Y) = ?$

$E[XY] = E[X X^2] = E[X^3] = 0$

$E[X] = 0$

$\mathrm{Cov}(X,Y) = 0$

X, Y are not independent

② $X = \begin{cases} +1 & \text{w.p. } 1/3 \\ 0 & 1/3 \\ -1 & 1/3 \end{cases} \qquad Y = X^2$

$Y = \begin{cases} 0 & 1/3 \\ 1 & 2/3 \end{cases}$

$E[X|Y] = E[X]$
doesn't imply X, Y are independent.

$E[X] = 0$ and $E[X|Y] = 0$

$E[Y] = 2/3 \qquad E[Y|X=x] = \begin{cases} 0 & x=0 \\ 1 & x=1 \\ 1 & x=-1 \end{cases}$

$E[X|Y=0] = \sum_{x \in \{-1 \ldots}$

$p_{X|Y}(x|0) = \{$

$p_{X|Y}(x|1) =$

$E[X|Y] = E[X]$

$\not\Downarrow$ doesn't imply.

$E[Y|X] = E[Y]$

$$E[X|Y=0] = \sum_{x \in \{-1,0,1\}} x\, P_{X|Y}(x|0) = 0$$

$$P_{X|Y}(x|0) = \begin{cases} 1 & x=0 \\ 0 & x=-1,1 \end{cases}$$

$$P_{X|Y}(x|1) = \frac{P_{X,Y}(x,1)}{P_Y(1)}$$

$X|Y=1 \sim \text{Uniform}(\{+1,-1\})$

$$\boxed{E[X|Y=1] = 0}$$

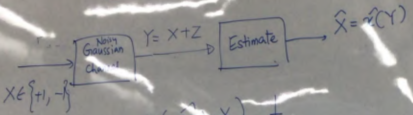
$$= \begin{cases} 0 & x=0 \\ \dfrac{P_{X,Y}(1,1)}{P_Y(1)} = \dfrac{1/3}{2/3} & x=1 \\ \dfrac{P_{X,Y}(-1,1)}{P_Y(1)} = \dfrac{1/3}{2/3} & x=-1 \end{cases} = \begin{cases} 1/2 & x \in \{+1,-1\} \\ 0 & x=0 \end{cases}$$

$$\boxed{\begin{gathered} E[X|Y] = E[X] \\ E[Y|X] = E[Y] \end{gathered}}$$

1/3

2/3

Lec 25



Prob of error $= P(\hat{X} \neq X) \downarrow$

$Z \sim N(0, \sigma^2)$ and Z is independent of X.

$\Downarrow$

$$f_{Z|X}(z|x) = f_Z(z).$$

$$X = \begin{cases} 1 & \text{w.p } 1/2 \\ -1 & \text{w.p } 1/2. \end{cases}$$

$$\hat{x}(y) = \begin{cases} 1 & y > 0 \\ -1 & y \leq 0. \end{cases}$$

$$P(X \neq \hat{X}) = P(X=1, \hat{X}=-1) + P(X=-1, \hat{X}=1)$$

$$= P(X=1)\, P(\hat{X}=-1 \mid X=1) + P(X=-1)\, P(\hat{X}=1 \mid X=-1)$$

$$= \frac{1}{2} P(Y \leq 0 \mid X=1) + \frac{1}{2} P(Y > 0 \mid X=-1)$$

$$= \frac{1}{2} P(X+Z \leq 0 \mid X=1) + \frac{1}{2} P(X+Z > 0 \mid X=-1)$$

$$= \frac{1}{2} P(Z \leq -1 \mid X=1) + \frac{1}{2} P(Z > 1 \mid X=-1)$$

$$= \frac{1}{2}\left[P(Z \leq -1) + P(Z > 1) \right]$$

$$= \frac{1}{2}\left[P(\sigma N \leq -1) + P(\sigma N > 1) \right]$$

Let N be standard normal R.V.

$Z = \sigma N$ is $N(0, \sigma^2)$ R.V.

$P(N \leq x) = \Phi(x)$

$1, \hat{X} = 1)$

$= 1 \mid X = -1)$

$P(Y > 0 \mid X = -1)$

$P(X + Z > 0 \mid X = -1)$

$P(Z > 1 \mid X = 1)$

$$= \frac{1}{2}\left[\Phi\left(\frac{-1}{\sigma}\right) + \left(1 - \Phi\left(\frac{1}{\sigma}\right)\right)\right] \qquad \Phi(-x) = 1 - \Phi(x)$$

$$= 1 - \Phi\left(\frac{1}{\sigma}\right)$$

$\sigma \uparrow \quad \frac{1}{\sigma} \downarrow \quad \Phi\left(\frac{1}{\sigma}\right) \downarrow \quad -\Phi\left(\frac{1}{\sigma}\right) \uparrow \quad 1 - \Phi\left(\frac{1}{\sigma}\right) \uparrow$

$Y \mid X = 1 \sim N(1, \sigma^2)$

$Y \mid X = -1 \sim N(-1, \sigma^2)$